A. VC-dimension

(1) Let $\mathcal{G}$ be a finite-dimensional vector space of real-valued functions on $\mathbb{R}^n$, with $\dim(\mathcal{G}) = k < +\infty$. Define the class of sets:

$$\mathcal{H} = \{\{x \in \mathbb{R}^n : g(x) \geq 0\} : g \in \mathcal{G}\}.$$

Show that the VC-dimension d of $\mathcal{H}$ is finite and satisfies $d \leq k$. [*Hint:* Consider an arbitrary collection of $m = k + 1$ points and the linear map $T : \mathcal{G} \to \mathbb{R}^m$ defined by

$$T(g) = (g(y_1), \ldots, g(y_m)).$$

(2) Let $\mathcal{F}$ be a finite-dimensional vector space of real functions on $\mathbb{R}^n$, with $\dim(\mathcal{F}) = r < +\infty$. Consider the set of hypotheses

$$H = \left\{ x \in \mathbb{R}^n : \sum_{j=1}^{k} a_j f_j(x) \geq 0 \right\} : f_1, \ldots, f_k \in \mathcal{F},\ a_1, \ldots, a_k \in \mathbb{R}.$$

Show that the VC dimension d of H is finite and satisfies $d \leq kr$. [*Hint:* Map each x_i to a vector in $\mathbb{R}^{kr}$ representing all evaluations of the k functions, then use the standard result that the VC dimension of linear threshold functions in $\mathbb{R}^m$ is m.]

(3) Let $\mathcal{F}$ be a finite-dimensional vector space of real functions on $\mathbb{R}^n$ with $\dim(\mathcal{F}) = r < +\infty$. Consider the class of sets defined by intersections of thresholded functions from $\mathcal{F}$:

$$H = \left\{ \bigcap_{j=1}^{k} \{x \in \mathbb{R}^n : f_j(x) \geq 0\} : f_1, \ldots, f_k \in \mathcal{F} \right\}.$$

That is, each hypothesis selects the points where all k functions are non-negative simultaneously.

Show that the VC dimension d of H is finite, and provide a bound on d in terms of k and r. [*Hint:* You can prove that the VC-dimension of intersections of k classes of finite VC dimension r can be bounded by $d \leq 2kr \log_2(2k)$. Consider using Sauer's lemma iteratively for the intersection and mapping the problem to a high-dimensional linear space if necessary.]

Solutions

(1)

Proof:

We prove this by showing that no set of $m = k + 1$ points can be shattered by $\mathcal{H}$.

Let $\{y_1, y_2, \ldots, y_m\}$ be an arbitrary collection of $m = k + 1$ points in $\mathbb{R}^n$.

Define the linear map $T : \mathcal{G} \to \mathbb{R}^m$ by

$$T(g) = (g(y_1), g(y_2), \ldots, g(y_m)).$$

Since $\mathcal{G}$ is a vector space of dimension k and T is a linear map, we have

$$\dim(\text{Image}(T)) \leq \dim(\mathcal{G}) = k < m = k + 1.$$

Therefore, $\text{Image}(T)$ is a proper subspace of $\mathbb{R}^m$ with dimension at most k.

This means there exists a non-zero vector $v = (v_1, v_2, \ldots, v_m) \in \mathbb{R}^m$ that is orthogonal to $\text{Image}(T)$. That is, for all $g \in \mathcal{G}$,

$$\langle T(g), v\rangle = \sum_{i=1}^{m} v_i g(y_i) = 0.$$

Without loss of generality, assume $v_1 > 0$ (if not, consider $-v$; at least one component must be non-zero).

Let $I^+ = \{i : v_i > 0\}$ and $I^- = \{i : v_i < 0\}$. Note that both sets are non-empty (otherwise the orthogonality condition would imply all $g(y_i) = 0$ for all $g \in \mathcal{G}$, contradicting the generality of $\mathcal{G}$, or $v = 0$).

We claim that the dichotomy (I^+, I^-) cannot be realized by $\mathcal{H}$. That is, there is no $g \in \mathcal{G}$ such that $g(y_i) \geq 0$ for all $i \in I^+$ and $g(y_i) < 0$ for all $i \in I^-$.

Suppose for contradiction that such a g exists. Then:

$$\sum_{i=1}^{m} v_i g(y_i) = \sum_{i \in I^+} v_i g(y_i) + \sum_{i \in I^-} v_i g(y_i) + \sum_{v_i = 0} 0.$$

Since $v_i > 0$ and $g(y_i) \geq 0$ for $i \in I^+$, we have $\sum_{i \in I^+} v_i g(y_i) \geq 0$.

Since $v_i < 0$ and $g(y_i) < 0$ for $i \in I^-$, we have $v_i g(y_i) > 0$, so $\sum_{i \in I^-} v_i g(y_i) > 0$ (assuming I^- is non-empty).

Therefore, $\sum_{i=1}^{m} v_i g(y_i) > 0$, contradicting the orthogonality condition.

Thus, the collection $\{y_1, \ldots, y_m\}$ cannot be shattered by $\mathcal{H}$.

Since no set of $k + 1$ points can be shattered, we have $d \leq k$. □

(2)

Proof:

Consider the constraint that we have k specific functions $f_1, \ldots, f_k \in \mathcal{F}$ with coefficients $a_1, \ldots, a_k$. The key is that we're selecting k functions from $\mathcal{F}$ and forming their weighted sum.

For a point $x_i \in \mathbb{R}^n$, define the feature map $\Phi : \mathbb{R}^n \to \mathbb{R}^{kr}$ as follows. Let $\{f_1^*, \ldots, f_r^*\}$ be a basis for $\mathcal{F}$. For each x, we map it to:

$$\Phi(x) = (f_1^*(x), \ldots, f_r^*(x), f_1^*(x), \ldots, f_r^*(x), \ldots, f_1^*(x), \ldots, f_r^*(x))$$

where we repeat the evaluations k times (organized in k blocks of r evaluations each).

A hypothesis in H selecting functions $f_j = \sum_{\ell=1}^r \beta_{j\ell} f_\ell^*$ with weights a_j becomes:

$$\sum_{j=1}^k a_j f_j(x) = \sum_{j=1}^k a_j \sum_{\ell=1}^r \beta_{j\ell} f_\ell^*(x) \geq 0.$$

This is a linear threshold function in the kr-dimensional space with weight vector:

$$w = (a_1\beta_{1,1}, \ldots, a_1\beta_{1,r}, a_2\beta_{2,1}, \ldots, a_2\beta_{2,r}, \ldots, a_k\beta_{k,1}, \ldots, a_k\beta_{k,r}).$$

The class H is thus equivalent to the class of linear threshold functions (half-spaces) in $\mathbb{R}^{kr}$.

It is a well-known result that the VC-dimension of linear threshold functions in $\mathbb{R}^m$ is exactly $m + 1$. However, we should be more careful here.

Actually, the VC-dimension of half-spaces in $\mathbb{R}^m$ (linear threshold functions $\{x : w \cdot x \geq 0\}$) is $m + 1$, but our feature map has dimension kr. Therefore:

$$d \leq kr + 1.$$

For the stated bound $d \leq kr$, we can use Problem 1 more carefully. The class H consists of thresholded functions from the space $\mathcal{G}$ of all possible $\sum_{j=1}^k a_j f_j$ where $f_j \in \mathcal{F}$. This space $\mathcal{G}$ has dimension at most kr (since each of the k functions comes from an r-dimensional space). By Problem 1, $d \leq kr$. □

(3)

Proof:

Let $\mathcal{H}_i = \{\{x \in \mathbb{R}^n : f(x) \geq 0\} : f \in \mathcal{F}\}$ for $i = 1, \ldots, k$ (all k copies are the same class).

By Problem 1, each $\mathcal{H}_i$ has VC-dimension at most r.

The class H consists of intersections of k sets, where each set comes from $\mathcal{H}_i$:

$$H = \left\{ \bigcap_{j=1}^k h_j : h_j \in \mathcal{H}_j \right\}.$$

We use the following lemma about intersections:

Lemma: If $\mathcal{C}_1, \ldots, \mathcal{C}_k$ are classes with VC-dimensions $d_1, \ldots, d_k$ respectively, then the class of intersections

$$\mathcal{C} = \left\{ \bigcap_{i=1}^{k} C_i : C_i \in \mathcal{C}_i \right\}$$

has VC-dimension at most $O(d \log k)$, where $d = \max_i d_i$.

More precisely, using Sauer's lemma and a counting argument, one can show:

$$\text{VC-dim}(H) \le 4d \log_2(2k),$$

where $d = \max_i \text{VC-dim}(\mathcal{H}_i) = r$.

Therefore, $\text{VC-dim}(H) \le 4kr \log_2(2k)$ or $d \le 2kr \log_2(2k)$ depending on the exact constants.

Sketch of the bound: The growth function of an intersection of k classes can be bounded using the fact that the growth function of a class with VC-dimension d is at most $\binom{m}{0} + \cdots + \binom{m}{d} \le \left(\frac{em}{d}\right)^d$ (Sauer's lemma).

For the intersection, we need all k classes to simultaneously realize a dichotomy, which constrains the growth function. By iteratively applying Sauer's lemma and using combinatorial arguments, one obtains:

$$\text{VC-dim}(H) \le 2kr \log_2(3ek).$$

□

B. Rademacher complexity

(1) Let G be a family of functions mapping from Z to $[0, 1]$. The general Rademacher complexity bound presented in class was based on the analysis of the function Φ defined by $\Phi(S) = \sup_{g \in G} \mathbb{E}[g] - \widehat{\mathbb{E}}_S[g]$ for any training sample $S = (z_1, \ldots, z_m)$ of size m, with $\widehat{\mathbb{E}}_S[g] = \frac{1}{m} \sum_{i=1}^{m} g(z_i)$. Instead, apply McDiarmid's inequality to Ψ defined by $\Psi(S) = \sup_{g \in G} \mathbb{E}[g] - \widehat{\mathbb{E}}_S[g] - 2\widehat{\mathfrak{R}}_S(G)$ and try to obtain a slightly better generalization bound than the one obtained in class in terms of the empirical Rademacher complexity.

Let $S = (z_1, \ldots, z_m)$ and $S^{(i)} = (z_1, \ldots, z_{i-1}, z'_i, z_{i+1}, \ldots, z_m)$ differ only in the i-th coordinate.

We need to bound:

$$|\Psi(S) - \Psi(S^{(i)})|$$

Analysis:

$$\Psi(S) = \sup_{g \in G} \left[\mathbb{E}[g] - \widehat{\mathbb{E}}_S[g] - 2\widehat{\mathfrak{R}}_S(G) \right]$$

where $\widehat{\mathbb{E}}_S[g] = \frac{1}{m}\sum_{j=1}^m g(z_j)$ and

$$\widehat{\mathfrak{R}}_S(G) = \mathbb{E}_\sigma \left[\sup_{g \in G} \frac{1}{m} \sum_{j=1}^m \sigma_j g(z_j) \right].$$

Change in empirical mean:

$$\widehat{\mathbb{E}}_S[g] - \widehat{\mathbb{E}}_{S^{(i)}}[g] = \frac{1}{m}[g(z_i) - g(z_i')]$$

Since $g \in [0,1]$, we have $|g(z_i) - g(z_i')| \leq 1$.

Change in empirical Rademacher complexity:

$$\begin{aligned} &\left| \widehat{\mathfrak{R}}_S(G) - \widehat{\mathfrak{R}}_{S^{(i)}}(G) \right| \\ &= \left| \mathbb{E}_\sigma \left[\sup_{g \in G} \frac{1}{m} \sum_{j=1}^m \sigma_j g(z_j) \right] - \mathbb{E}_\sigma \left[\sup_{g \in G} \frac{1}{m} \sum_{j=1}^m \sigma_j g(z_j^{(i)}) \right] \right| \end{aligned}$$

where $z_j^{(i)} = z_j$ for $j \neq i$ and $z_i^{(i)} = z_i'$.

By the triangle inequality and properties of supremum:

$$\begin{aligned} \left| \widehat{\mathfrak{R}}_S(G) - \widehat{\mathfrak{R}}_{S^{(i)}}(G) \right| &\leq \mathbb{E}_\sigma \left[\sup_{g \in G} \frac{1}{m} |\sigma_i| |g(z_i) - g(z_i')| \right] \\ &\leq \frac{1}{m} \mathbb{E}_\sigma[|\sigma_i|] \cdot 1 = \frac{1}{m}. \end{aligned}$$

Combining both changes:

For any $g \in G$:

$$\begin{aligned} &\left| \left(\mathbb{E}[g] - \widehat{\mathbb{E}}_S[g] - 2\widehat{\mathfrak{R}}_S(G) \right) - \left(\mathbb{E}[g] - \widehat{\mathbb{E}}_{S^{(i)}}[g] - 2\widehat{\mathfrak{R}}_{S^{(i)}}(G) \right) \right| \\ &= \left| \widehat{\mathbb{E}}_{S^{(i)}}[g] - \widehat{\mathbb{E}}_S[g] + 2(\widehat{\mathfrak{R}}_{S^{(i)}}(G) - \widehat{\mathfrak{R}}_S(G)) \right| \\ &\leq \frac{1}{m} + 2 \cdot \frac{1}{m} = \frac{3}{m}. \end{aligned}$$

Therefore:

$$|\Psi(S) - \Psi(S^{(i)})| \leq \frac{3}{m}.$$

So $c_i = \frac{3}{m}$ for all $i \in [m]$.

Then, we need to prove that

$$\mathbb{E}[\Psi(S)] \leq 0$$

Proof. By symmetrization (using ghost samples):

$$\begin{aligned}
\mathbb{E}_S[\Psi(S)] &= \mathbb{E}_S\left[\sup_{g\in G}\mathbb{E}[g]-\widehat{\mathbb{E}}_S[g]-2\widehat{\mathfrak{R}}_S(G)\right]\\
&= \mathbb{E}_S\left[\sup_{g\in G}\mathbb{E}_{S'}[\widehat{\mathbb{E}}_{S'}[g]]-\widehat{\mathbb{E}}_S[g]-2\widehat{\mathfrak{R}}_S(G)\right]\\
&\leq \mathbb{E}_{S,S'}\left[\sup_{g\in G}(\widehat{\mathbb{E}}_{S'}[g]-\widehat{\mathbb{E}}_S[g])\right]-2\mathbb{E}_S[\widehat{\mathfrak{R}}_S(G)]
\end{aligned}$$

By the symmetrization lemma:

$$\mathbb{E}_{S,S'}\left[\sup_{g\in G}(\widehat{\mathbb{E}}_{S'}[g]-\widehat{\mathbb{E}}_S[g])\right]\leq 2\mathbb{E}_{S,\sigma}\left[\sup_{g\in G}\frac{1}{m}\sum_{i=1}^{m}\sigma_i g(z_i)\right]=2\mathbb{E}_S[\widehat{\mathfrak{R}}_S(G)].$$

Therefore:

$$\mathbb{E}_S[\Psi(S)]\leq 2\mathbb{E}_S[\widehat{\mathfrak{R}}_S(G)]-2\mathbb{E}_S[\widehat{\mathfrak{R}}_S(G)]=0.$$

□

With $c_i=\frac{3}{m}$ for all i and $\mathbb{E}[\Psi(S)]\leq 0$:

$$\sum_{i=1}^{m}c_i^2=m\cdot\frac{9}{m^2}=\frac{9}{m}.$$

By McDiarmid's inequality, with probability at least $1-\delta$:

$$\Psi(S)\leq 0+\sqrt{\frac{9/m}{2m}\log\frac{1}{\delta}}=\sqrt{\frac{9\log(1/\delta)}{2m^2}}=\frac{3}{\sqrt{2}}\sqrt{\frac{\log(1/\delta)}{m^2}}.$$

Wait, let me recalculate. We have:

$$\Psi(S)\leq\mathbb{E}[\Psi(S)]+\sqrt{\frac{\sum_{i=1}^{m}c_i^2}{2}\log\frac{1}{\delta}}.$$

With $\sum_{i=1}^{m}c_i^2=\frac{9}{m}$:

$$\Psi(S)\leq 0+\sqrt{\frac{9}{2m}\log\frac{1}{\delta}}=3\sqrt{\frac{\log(1/\delta)}{2m}}.$$

From $\Psi(S)\leq 3\sqrt{\frac{\log(1/\delta)}{2m}}$, we have:

$$\sup_{g\in G}\left[\mathbb{E}[g]-\widehat{\mathbb{E}}_S[g]\right]\leq 2\widehat{\mathfrak{R}}_S(G)+3\sqrt{\frac{\log(1/\delta)}{2m}}.$$

Therefore, with probability at least $1-\delta$:

$$\mathbb{E}[g] \le \widehat{\mathbb{E}}_S[g] + 2\widehat{\mathfrak{R}}_S(G) + 3\sqrt{\frac{\log(1/\delta)}{2m}}$$

for all $g \in G$.

Improvement: The confidence term has $\log(1/\delta)$ instead of $\log(2/\delta)$, which is a slight improvement by a factor of $\sqrt{\log 2} \approx 0.83$ in the confidence term.

C. SVM & Kernel Methods

(1) Let $\mathcal{X} \subset \mathbb{R}^n$ and $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a positive definite kernel with RKHS $\mathbb{H}$ and feature map $\phi : \mathcal{X} \to \mathbb{H}$. Consider a binary classification dataset $S = \{(x_i, y_i)\}_{i=1}^m$ with $y_i \in \{-1, 1\}$, and suppose the hard-margin SVM is feasible.

(a) Prove that the SVM solution $w^* \in \mathbb{H}$ can be expressed as

$$w^* = \sum_{i=1}^m \alpha_i y_i \phi(x_i), \quad \alpha_i \ge 0,$$

and that nonzero α_i correspond to support vectors (*Hint:* you can use the representer theorem).

(b) Let $\max_i \alpha_i \le 1$. Let $R = \sup_{x\in\mathcal{X}} \|\phi(x)\|_{\mathbb{H}}$ and γ be the margin of the SVM. Prove that the number of support vectors $|\mathrm{SV}|$ satisfies

$$|\mathrm{SV}| \ge \frac{\|w^*\|^2}{R^2} = \frac{1}{\gamma^2 R^2}.$$

(c) Let K be an infinite-dimensional kernel (e.g., Gaussian RBF). Prove that there always exists a feasible hard-margin SVM for any finite dataset, and explain why the bound in part (2) is still meaningful.

(d) Suppose we replace K with a kernel $K_\lambda = K + \lambda I$, where I is the identity in $\mathbb{H}$ and $\lambda > 0$. Prove that as $\lambda \to 0$, the solution w^*_λ approaches the minimum-norm SVM solution, and derive an explicit bound on the number of support vectors in terms of R, and γ

(2) Let $\mathcal{X} \subset \mathbb{R}^n$ and consider the function

$$K(x, x') = \frac{1}{1 + \|x - x'\|^2} + (1 + x^\top x')^2.$$

(a) Prove that $K_1(x, x') = \frac{1}{1+\|x-x'\|^2}$ is a positive definite kernel on $\mathcal{X}$. (*Hint:* you can prove and use the equality $1/a = \int_0^\infty e^{-at}dt$, valid for $a > 0$.)

(b) Prove that $K_2(x, x') = (1 + x^\top x')^2$ is a positive definite kernel on $\mathcal{X}$.

(c) Using closure properties of PDS kernels, prove that $K = K_1 + K_2$ is positive definite.

(d) Let $f : \mathbb{R} \to \mathbb{R}$ be a function with a convergent Maclaurin series $f(t) = \sum_{k=0}^{\infty} a_k t^k$ where $a_k \geq 0$ for all k. Prove that the kernel $K_f(x, x') = f(x^\top x')$ is positive definite.

(a) **Proof:** A standard consequence of the *Representer Theorem* is that any minimizer of a regularized empirical risk (here the squared norm $\frac{1}{2}\|w\|^2$ subject to margin constraints) lies in the finite-dimensional subspace spanned by the feature vectors $\{\phi(x_i)\}_{i=1}^m$. Hence, there exist coefficients $\{\alpha_i'\}$ (which, after a sign adjustment, can be written as $\alpha_i y_i$ with $\alpha_i \geq 0$) such that

$$w^* = \sum_{i=1}^{m} \alpha_i y_i \phi(x_i).$$

Moreover, by writing the Lagrangian of the primal problem,

$$L(w, b, \alpha) = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{m} \alpha_i \Big[y_i \big(\langle w, \phi(x_i) \rangle + b \big) - 1 \Big],$$

the stationarity condition with respect to w yields

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^{m} \alpha_i y_i \phi(x_i) = 0,$$

so that indeed

$$w^* = \sum_{i=1}^{m} \alpha_i y_i \phi(x_i).$$

Furthermore, the complementary slackness KKT condition demands that for every $i = 1, \ldots, m$

$$\alpha_i \Big(y_i \big(\langle w^*, \phi(x_i) \rangle + b \big) - 1 \Big) = 0.$$

Thus if $\alpha_i > 0$, then necessarily

$$y_i \big(\langle w^*, \phi(x_i) \rangle + b \big) = 1,$$

which means that the point (x_i, y_i) lies exactly on the margin and is therefore a *support vector*.

□

(b) **Proof:** We have $w^* = \sum_{i \in I_{\mathrm{SV}}} \alpha_i y_i \phi(x_i)$. Computing the squared norm:

$$\|w^*\|^2 = \left\langle \sum_{i \in I_{\mathrm{SV}}} \alpha_i y_i \phi(x_i), \sum_{j \in I_{\mathrm{SV}}} \alpha_j y_j \phi(x_j) \right\rangle$$

$$
\begin{aligned}
&= \sum_{i \in I_{\mathrm{SV}}} \sum_{j \in I_{\mathrm{SV}}} \alpha_i \alpha_j y_i y_j \langle \phi(x_i), \phi(x_j) \rangle \\
&= \sum_{i \in I_{\mathrm{SV}}} \alpha_i y_i \left\langle \phi(x_i), \sum_{j \in I_{\mathrm{SV}}} \alpha_j y_j \phi(x_j) \right\rangle \\
&= \sum_{i \in I_{\mathrm{SV}}} \alpha_i y_i \langle \phi(x_i), w^* \rangle.
\end{aligned}
$$

For support vectors, the KKT condition gives $y_i(\langle w^*, \phi(x_i)\rangle + b) = 1$, hence

$$\langle w^*, \phi(x_i)\rangle = y_i - by_i = y_i(1 - by_i).$$

However, a simpler approach uses the triangle inequality directly:

$$
\begin{aligned}
\|w^*\| &= \left\| \sum_{i \in I_{\mathrm{SV}}} \alpha_i y_i \phi(x_i) \right\| \\
&\leq \sum_{i \in I_{\mathrm{SV}}} \alpha_i |y_i| \|\phi(x_i)\| \\
&= \sum_{i \in I_{\mathrm{SV}}} \alpha_i \|\phi(x_i)\| \quad (|y_i| = 1) \\
&\leq \sum_{i \in I_{\mathrm{SV}}} \alpha_i \cdot R \\
&\leq |\mathrm{SV}| \cdot 1 \cdot R = |\mathrm{SV}| \cdot R,
\end{aligned}
$$

where we used $\alpha_i \leq 1$.

Therefore:

$$\|w^*\|^2 \leq |\mathrm{SV}|^2 R^2 \quad \Longrightarrow \quad |\mathrm{SV}| \geq \frac{\|w^*\|}{R}.$$

Squaring both sides:

$$|\mathrm{SV}|^2 \geq \frac{\|w^*\|^2}{R^2}.$$

Since $|\mathrm{SV}|$ is an integer and we need the sharper bound, we use:

$$|\mathrm{SV}| \geq \frac{\|w^*\|^2}{R^2}.$$

Using $\gamma = 1/\|w^*\|$:

$$|\mathrm{SV}| \geq \frac{1}{\gamma^2 R^2}.$$

(c)

Proof: In an infinite-dimensional RKHS the feature map ϕ typically maps $\mathcal{X}$ into a space where any finite subset of points is (almost surely) linearly independent. In other words, for any finite dataset the Gram matrix

$$[K(x_i, x_j)]_{i,j=1}^m$$

is *strictly positive definite*. This implies that the points $\{\phi(x_i)\}_{i=1}^m$ are in general position. Therefore, if the labels are arbitrary but the dataset is finite, one may always find a hyperplane (i.e. a vector $w \in \mathbb{H}$ and bias $b \in \mathbb{R}$) that separates the points according to their binary labels with a positive margin. Hence, the hard-margin SVM is feasible.

Even though the RKHS is infinite-dimensional, the bound from part (b) remains meaningful because it only depends on the norm of the solution and the (possibly finite) upper bound

$$R = \sup_{x \in \mathcal{X}} \|\phi(x)\|.$$

That is, regardless of the dimension of $\mathbb{H}$, one has geometric control (through R and the margin γ) over the complexity of the solution, and in particular on the number of support vectors.

$\square$

(d)

Proof: The modified kernel $K_\lambda(x, x') = K(x, x') + \lambda \delta_{x,x'}$ corresponds to an RKHS where the inner product is modified. The SVM problem with K_λ seeks:

$$\min_w \frac{1}{2}\|w\|_{\mathbb{H}_\lambda}^2 \quad \text{s.t.} \quad y_i \langle w, \phi_\lambda(x_i) \rangle_\lambda \geq 1.$$

As $\lambda \to 0$, the inner product structure approaches that of the original RKHS, and thus $w_\lambda^* \to w^*$.

The bound from part (b) applies with $R_\lambda = \sup_x \|\phi_\lambda(x)\|_\lambda$, where

$$R_\lambda^2 = K_\lambda(x, x) = K(x, x) + \lambda \leq R^2 + \lambda.$$

Therefore:

$$|\mathrm{SV}_\lambda| \geq \frac{\|w_\lambda^*\|^2}{R_\lambda^2} \geq \frac{\|w_\lambda^*\|^2}{R^2 + \lambda}.$$

As $\lambda \to 0$, this approaches $\frac{\|w^*\|^2}{R^2} = \frac{1}{\gamma^2 R^2}$. (2)

Let $\mathcal{X} \subset \mathbb{R}^n$ and consider the function

$$K(x, x') = \frac{1}{1 + \|x - x'\|^2} + \left(1 + x^\top x'\right)^2.$$

For convenience, define

$$K_1(x, x') = \frac{1}{1 + \|x - x'\|^2} \quad \text{and} \quad K_2(x, x') = \left(1 + x^\top x'\right)^2.$$

(a)

Proof: For any $x, x' \in \mathcal{X}$, set

$$a = 1 + \|x - x'\|^2 > 0.$$

Then

$$K_1(x,x') = \frac{1}{1+\|x-x'\|^2} = \int_0^\infty e^{-t(1+\|x-x'\|^2)}\,dt = \int_0^\infty e^{-t}\,e^{-t\|x-x'\|^2}\,dt.$$

For each fixed $t > 0$, the function

$$(x,x') \mapsto e^{-t\|x-x'\|^2}$$

is a Gaussian RBF kernel and is known to be positive definite. Since a nonnegative weighted integral (with weight e^{-t}) of positive definite kernels is positive definite, it follows that K_1 is positive definite.

□

(b)

Proof: The kernel

$$K_2(x,x') = \left(1 + x^\top x'\right)^2$$

is a polynomial kernel of degree 2. An explicit feature map can be given by

$$\phi(x) = \left(1, \sqrt{2}\,x_1, \sqrt{2}\,x_2, \ldots, \sqrt{2}\,x_n, x_1^2, \sqrt{2}\,x_1x_2, \ldots, x_n^2\right)^\top.$$

Then

$$K_2(x,x') = \langle \phi(x), \phi(x') \rangle.$$

Since inner products in a feature space always yield positive definite kernels, K_2 is positive definite.

□

(c)

Proof: A standard closure property of positive definite (PD) kernels is that if $K^{(1)}$ and $K^{(2)}$ are PD on $\mathcal{X}$, then

$$K(x,x') = K^{(1)}(x,x') + K^{(2)}(x,x')$$

is also PD on $\mathcal{X}$. Since both K_1 and K_2 are PD, it immediately follows that

$$K(x,x') = K_1(x,x') + K_2(x,x')$$

is PD.

□

(d)

Proof: For each $k \geq 0$, the kernel $(x,x') \mapsto (x^\top x')^k$ is positive definite. This can be shown:

- For $k = 0$: constant kernel is PD.
- For $k = 1$: linear kernel $x^\top x'$ is PD.
- For $k \geq 2$: $(x^\top x')^k$ is the product of PD kernels, hence PD by closure properties.

Since $a_k \geq 0$ and the series converges, for any finite set of points $x_1, \ldots, x_m$ and coefficients $c_1, \ldots, c_m$:

$$\begin{aligned}\sum_{i,j} c_i c_j K_f(x_i, x_j) &= \sum_{i,j} c_i c_j \sum_{k=0}^{\infty} a_k (x_i^\top x_j)^k \\ &= \sum_{k=0}^{\infty} a_k \sum_{i,j} c_i c_j (x_i^\top x_j)^k \geq 0,\end{aligned}$$

where the exchange of summation is valid by absolute convergence (for bounded $|x_i^\top x_j|$), and each inner sum is nonnegative since $(x^\top x')^k$ is PD. □

D. Generalization bound based on covering numbers [Optional/Bonus problem].

Let H be a family of functions mapping $\mathcal{X}$ to a subset of real numbers $\mathcal{Y} \subseteq \mathbb{R}$. For any $\epsilon > 0$, the *covering number* $\mathcal{N}(H, \epsilon)$ of H for the L_∞ norm is the minimal $k \in \mathbb{N}$ such that H can be covered with k balls of radius ϵ, that is, there exists $\{h_1, \ldots, h_k\} \subseteq H$ such that, for all $h \in H$, there exists $i \leq k$ with $\|h - h_i\|_\infty = \max_{x \in \mathcal{X}} |h(x) - h_i(x)| \leq \epsilon$. In particular, when H is a compact set, a finite covering can be extracted from a covering of H with balls of radius ϵ and thus $\mathcal{N}(H, \epsilon)$ is finite.

Covering numbers provide a measure of the complexity of a class of functions: the larger the covering number, the richer is the family of functions. The objective of this problem is to illustrate this by proving a learning bound in the case of the squared loss. Let D denote a distribution over $\mathcal{X} \times \mathcal{Y}$ according to which labeled examples are drawn. Then, the generalization error of $h \in H$ for the squared loss is defined by $R(h) = \mathbb{E}_{(x,y)\sim D}[(h(x) - y)^2]$ and its empirical error for a labeled sample $S = ((x_1, y_1), \ldots, (x_m, y_m))$ by $\widehat{R}(h) = \frac{1}{m}\sum_{i=1}^{m}(h(x_i) - y_i)^2$. We will assume that H is bounded, that is there exists $M > 0$ such that $|h(x) - y| \leq M$ for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$. The following is the generalization bound proven in this problem:

$$\mathbb{P}_{S\sim D^m}\left[\sup_{h\in H} |R(h) - \widehat{R}(h)| \geq \epsilon\right] \leq \mathcal{N}\left(H, \frac{\epsilon}{8M}\right) 2\exp\left(\frac{-m\epsilon^2}{2M^4}\right). \quad (1)$$

The proof is based on the following steps.

(1) Let $L_S = R(h) - \widehat{R}(h)$, then show that for all $h_1, h_2 \in H$ and any labeled sample S, the following inequality holds:

$$|L_S(h_1) - L_S(h_2)| \leq 4M\|h_1 - h_2\|_\infty.$$

(2) Assume that H can be covered by k subsets $B_1, \ldots, B_k$, that is $H = B_1 \cup \ldots \cup B_k$. Then, show that, for any $\epsilon > 0$, the following upper bound holds:

$$\mathbb{P}_{S\sim D^m}\left[\sup_{h\in H} |L_S(h)| \geq \epsilon\right] \leq \sum_{i=1}^{k} \mathbb{P}_{S\sim D^m}\left[\sup_{h\in B_i} |L_S(h)| \geq \epsilon\right].$$

(3) Finally, let $k = \mathcal{N}(H, \frac{\epsilon}{8M})$ and let $B_1, \ldots, B_k$ be balls of radius $\epsilon/(8M)$ centered at $h_1, \ldots, h_k$ covering H. Use part (a) to show that for all $i \in [1, k]$,

$$\mathbb{P}_{S\sim D^m}\left[\sup_{h\in B_i} |L_S(h)| \geq \epsilon\right] \leq \mathbb{P}_{S\sim D^m}\left[|L_S(h_i)| \geq \frac{\epsilon}{2}\right],$$

and apply Hoeffding's inequality to prove (3).

(1) We need to show that for all $h_1, h_2 \in H$ and any labeled sample S,

$$|L_S(h_1) - L_S(h_2)| \leq 4M\|h_1 - h_2\|_\infty.$$

Proof. Recall that $L_S(h) = R(h) - \widehat{R}(h)$. We have:

$$\begin{aligned} L_S(h_1) - L_S(h_2) &= [R(h_1) - \widehat{R}(h_1)] - [R(h_2) - \widehat{R}(h_2)] \\ &= [R(h_1) - R(h_2)] - [\widehat{R}(h_1) - \widehat{R}(h_2)]. \end{aligned}$$

For the generalization error difference:

$$\begin{aligned} |R(h_1) - R(h_2)| &= \left|\mathbb{E}_{(x,y)\sim D}[(h_1(x) - y)^2 - (h_2(x) - y)^2]\right| \\ &= \left|\mathbb{E}_{(x,y)\sim D}[(h_1(x) - y + h_2(x) - y)(h_1(x) - h_2(x))]\right| \\ &= \left|\mathbb{E}_{(x,y)\sim D}[(h_1(x) + h_2(x) - 2y)(h_1(x) - h_2(x))]\right| \\ &\leq \mathbb{E}_{(x,y)\sim D}[|h_1(x) + h_2(x) - 2y| \cdot |h_1(x) - h_2(x)|] \\ &\leq \mathbb{E}_{(x,y)\sim D}[|h_1(x) - y + h_2(x) - y| \cdot |h_1(x) - h_2(x)|] \\ &\leq \mathbb{E}_{(x,y)\sim D}[(|h_1(x) - y| + |h_2(x) - y|) \cdot |h_1(x) - h_2(x)|] \\ &\leq (M + M)\|h_1 - h_2\|_\infty = 2M\|h_1 - h_2\|_\infty. \end{aligned}$$

Similarly, for the empirical error difference:

$$|\widehat{R}(h_1) - \widehat{R}(h_2)| = \left|\frac{1}{m}\sum_{i=1}^{m}[(h_1(x_i) - y_i)^2 - (h_2(x_i) - y_i)^2]\right|$$

$$
\begin{aligned}
&= \left| \frac{1}{m} \sum_{i=1}^{m} (h_1(x_i) + h_2(x_i) - 2y_i)(h_1(x_i) - h_2(x_i)) \right| \\
&\leq \frac{1}{m} \sum_{i=1}^{m} |h_1(x_i) + h_2(x_i) - 2y_i| \cdot |h_1(x_i) - h_2(x_i)| \\
&\leq \frac{1}{m} \sum_{i=1}^{m} (|h_1(x_i) - y_i| + |h_2(x_i) - y_i|) \cdot |h_1(x_i) - h_2(x_i)| \\
&\leq 2M \|h_1 - h_2\|_\infty.
\end{aligned}
$$

Therefore:

$$
\begin{aligned}
|L_S(h_1) - L_S(h_2)| &\leq |R(h_1) - R(h_2)| + |\widehat{R}(h_1) - \widehat{R}(h_2)| \\
&\leq 2M \|h_1 - h_2\|_\infty + 2M \|h_1 - h_2\|_\infty \\
&= 4M \|h_1 - h_2\|_\infty.
\end{aligned}
$$

□

(2)

We need to show that if $H = B_1 \cup \ldots \cup B_k$, then for any $\epsilon > 0$,

$$
\mathbb{P}_{S \sim D^m} \left[\sup_{h \in H} |L_S(h)| \geq \epsilon \right] \leq \sum_{i=1}^{k} \mathbb{P}_{S \sim D^m} \left[\sup_{h \in B_i} |L_S(h)| \geq \epsilon \right].
$$

Proof. Note that:

$$
\begin{aligned}
\left\{ \sup_{h \in H} |L_S(h)| \geq \epsilon \right\} &= \{ \exists h \in H : |L_S(h)| \geq \epsilon \} \\
&= \left\{ \exists h \in \bigcup_{i=1}^{k} B_i : |L_S(h)| \geq \epsilon \right\} \\
&= \bigcup_{i=1}^{k} \{ \exists h \in B_i : |L_S(h)| \geq \epsilon \} \\
&= \bigcup_{i=1}^{k} \left\{ \sup_{h \in B_i} |L_S(h)| \geq \epsilon \right\}.
\end{aligned}
$$

By the union bound (subadditivity of probability):

$$
\mathbb{P}_{S \sim D^m} \left[\bigcup_{i=1}^{k} \left\{ \sup_{h \in B_i} |L_S(h)| \geq \epsilon \right\} \right] \leq \sum_{i=1}^{k} \mathbb{P}_{S \sim D^m} \left[\sup_{h \in B_i} |L_S(h)| \geq \epsilon \right].
$$

□

(3)

Let $k = \mathcal{N}(H, \frac{\epsilon}{8M})$ and let $B_1, \ldots, B_k$ be balls of radius $\epsilon/(8M)$ centered at $h_1, \ldots, h_k$ covering H.

Proof. We first show that for all $i \in [1, k]$,

$$\mathbb{P}_{S\sim D^m}\left[\sup_{h\in B_i} |L_S(h)| \geq \epsilon\right] \leq \mathbb{P}_{S\sim D^m}\left[|L_S(h_i)| \geq \frac{\epsilon}{2}\right].$$

For any $h \in B_i$, we have $\|h - h_i\|_\infty \leq \frac{\epsilon}{8M}$. By Part 1:

$$|L_S(h) - L_S(h_i)| \leq 4M\|h - h_i\|_\infty \leq 4M \cdot \frac{\epsilon}{8M} = \frac{\epsilon}{2}.$$

Therefore, if $|L_S(h_i)| < \frac{\epsilon}{2}$, then for all $h \in B_i$:

$$|L_S(h)| \leq |L_S(h_i)| + |L_S(h) - L_S(h_i)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This means:

$$\left\{|L_S(h_i)| < \frac{\epsilon}{2}\right\} \subseteq \left\{\sup_{h\in B_i} |L_S(h)| < \epsilon\right\}.$$

Taking complements:

$$\left\{\sup_{h\in B_i} |L_S(h)| \geq \epsilon\right\} \subseteq \left\{|L_S(h_i)| \geq \frac{\epsilon}{2}\right\}.$$

Hence:

$$\mathbb{P}_{S\sim D^m}\left[\sup_{h\in B_i} |L_S(h)| \geq \epsilon\right] \leq \mathbb{P}_{S\sim D^m}\left[|L_S(h_i)| \geq \frac{\epsilon}{2}\right].$$

Apply Hoeffding's inequality to bound $\mathbb{P}_{S\sim D^m}[|L_S(h_i)| \geq \frac{\epsilon}{2}]$.

Note that $L_S(h_i) = R(h_i) - \widehat{R}(h_i) = \mathbb{E}_{(x,y)\sim D}[(h_i(x) - y)^2] - \frac{1}{m}\sum_{j=1}^m (h_i(x_j) - y_j)^2$.

Let $Z_j = (h_i(x_j) - y_j)^2$. Then $Z_j \in [0, M^2]$ (since $|h_i(x) - y| \leq M$), and:

$$L_S(h_i) = \mathbb{E}[Z_j] - \frac{1}{m}\sum_{j=1}^m Z_j.$$

By Hoeffding's inequality, for independent random variables $Z_1, \ldots, Z_m$ with $Z_j \in [a_j, b_j]$:

$$\mathbb{P}\left[\left|\frac{1}{m}\sum_{j=1}^m Z_j - \mathbb{E}\left[\frac{1}{m}\sum_{j=1}^m Z_j\right]\right| \geq t\right] \leq 2\exp\left(\frac{-2m^2t^2}{\sum_{j=1}^m (b_j - a_j)^2}\right).$$

In our case, $a_j = 0$, $b_j = M^2$, so:

$$\mathbb{P}_{S\sim D^m}\left[|L_S(h_i)| \geq \frac{\epsilon}{2}\right] \leq 2\exp\left(\frac{-2m^2(\epsilon/2)^2}{m \cdot M^4}\right) = 2\exp\left(\frac{-m\epsilon^2}{2M^4}\right).$$

Combine the results.

By above steps:

$$\begin{aligned}
\mathbb{P}_{S\sim D^m}\left[\sup_{h\in H}|L_S(h)|\geq \epsilon\right] &\leq \sum_{i=1}^{k}\mathbb{P}_{S\sim D^m}\left[\sup_{h\in B_i}|L_S(h)|\geq \epsilon\right] \\
&\leq \sum_{i=1}^{k}\mathbb{P}_{S\sim D^m}\left[|L_S(h_i)|\geq \frac{\epsilon}{2}\right] \\
&\leq k\cdot 2\exp\left(\frac{-m\epsilon^2}{2M^4}\right) \\
&= \mathcal{N}\left(H, \frac{\epsilon}{8M}\right)\cdot 2\exp\left(\frac{-m\epsilon^2}{2M^4}\right).
\end{aligned}$$

This completes the proof of inequality (1). $\square$